

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 May/June 2022 Worked Solutions

May/June 2022 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 May/June 2022

May/June 2022 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Binomial Expansion

1. Find the first four terms, in ascending powers of x , of the binomial expansion of

$$\left(2 + \frac{3}{8}x\right)^{10}$$

Give each coefficient as an integer.

(4)

Worked Solution - Question 1

1. Write the binomial terms

$$\left(2 + \frac{3}{8}x\right)^{10} = 2^{10} + \binom{10}{1}2^9\left(\frac{3}{8}x\right) + \binom{10}{2}2^8\left(\frac{3}{8}x\right)^2 + \binom{10}{3}2^7\left(\frac{3}{8}x\right)^3 + \dots$$

2. Simplify the constant and linear terms

$$2^{10} = 1024 \text{ and } \binom{10}{1}2^9\left(\frac{3}{8}\right)x = 1920x.$$

3. Simplify the quadratic term

$$\binom{10}{2}2^8\left(\frac{3}{8}\right)^2x^2 = 1620x^2.$$

4. Simplify the cubic term

$$\binom{10}{3}2^7\left(\frac{3}{8}\right)^3x^3 = 810x^3.$$

5. State the expansion

The first four terms are $1024 + 1920x + 1620x^2 + 810x^3$.

Final answer

$$1024 + 1920x + 1620x^2 + 810x^3.$$

Question 2

Integration

2.

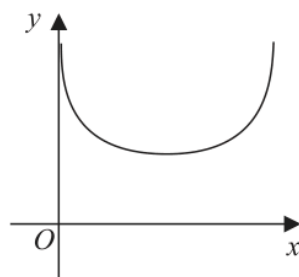


Figure 1

Figure 1 shows the graph of

$$y = 1 - \log_{10}(\sin x) \quad 0 < x < \pi$$

where x is in radians.

The table below shows some values of x and y for this graph, with values of y given to 3 decimal places.

x	0.5	1	1.5	2	2.5	3
y	1.319		1.001		1.223	1.850

- (a) Complete the table above, giving values of y to 3 decimal places. (2)
- (b) Use the trapezium rule with all the y values in the completed table to find, to 2 decimal places, an estimate for

$$\int_{0.5}^3 (1 - \log_{10}(\sin x)) dx \quad (3)$$

- (c) Use your answer to part (b) to find an estimate for

$$\int_{0.5}^3 (3 + \log_{10}(\sin x)) dx \quad (3)$$

Worked Solution - Question 2

1. Complete the missing table values

Using $y = 1 - \log_{10}(\sin x)$ gives $y(1) = 1.075$ and $y(2) = 1.041$ to 3 decimal places.

2. Identify the strip width

The x values increase by 0.5, so $h = 0.5$.

3. Apply the trapezium rule

$$A \approx \frac{0.5}{2} \{1.319 + 1.850 + 2(1.075 + 1.001 + 1.041 + 1.223)\}.$$

4. Calculate part b

This gives $2.9625 \dots$, so the estimate is **2.96** to 2 decimal places.

5. Rewrite the new integrand

Since $y = 1 - \log_{10}(\sin x)$, we have $\log_{10}(\sin x) = 1 - y$.

6. Use the interval length

$3 + \log_{10}(\sin x) = 4 - y$, and the interval from 0.5 to 3 has length 2.5.

7. Estimate part c

$$\int_{0.5}^3 (3 + \log_{10}(\sin x)) dx \approx 4(2.5) - 2.9625 \dots = 7.0374 \dots, \text{ so } 7.04.$$

Final answer

(a) $y(1) = 1.075$, $y(2) = 1.041$. (b) 2.96. (c) 7.04.

Question 3

Proof

3. (i) Show that the following statement is **false**:

“($n + 1$)³ − n ³ is prime for all $n \in \mathbb{N}$ ”

(2)

- (ii) Given that the points $A(1, 0)$, $B(3, -10)$ and $C(7, -6)$ lie on a circle, prove that AB is a diameter of this circle.

(5)

Worked Solution - Question 3

Topic group

1. Choose a counterexample

Take $n = 5$.

2. Evaluate the expression

$$(5 + 1)^3 - 5^3 = 216 - 125 = 91.$$

3. Show it is not prime

$91 = 7 \times 13$, so the statement is false.

4. Find the gradient CA

For $A(1, 0)$ and $C(7, -6)$, the gradient of CA is $\frac{0 - (-6)}{1 - 7} = -1$.

5. Find the gradient CB

For $B(3, -10)$ and $C(7, -6)$, the gradient of CB is $\frac{-10 - (-6)}{3 - 7} = 1$.

6. Show the right angle

The product of the gradients is -1 , so CA is perpendicular to CB and $\angle ACB = 90^\circ$.

7. Conclude the diameter

An angle of 90° at the circumference subtends a diameter, so AB is a diameter of the circle.

Final answer

(i) Counterexample: $n = 5$ gives **91**, not prime.

(ii) Proven: $\overline{AB}$ is a diameter.

Question 4

Laws of Logarithms

4. In this question you must show all stages of your working.

Give your answers in fully simplified surd form.

Given that a and b are positive constants, solve the simultaneous equations

$$\begin{aligned}a - b &= 8 \\ \log_4 a + \log_4 b &= 3\end{aligned}$$

(6)

Worked Solution - Question 4

Topic group

1. Combine the logarithms

$$\log_4 a + \log_4 b = \log_4(ab) = 3.$$

2. Find ab

$$\text{Therefore } ab = 4^3 = 64.$$

3. Use a minus b

$$\text{From } a - b = 8, b = a - 8.$$

4. Form a quadratic

$$a(a - 8) = 64, \text{ so } a^2 - 8a - 64 = 0.$$

5. Solve for a

$$a = \frac{8 \pm \sqrt{64 + 256}}{2} = 4 \pm 4\sqrt{5}.$$

6. Choose positive constants

$$\text{Since } b = a - 8 \text{ must also be positive, take } a = 4 + 4\sqrt{5}.$$

7. Find b

$$b = a - 8 = 4\sqrt{5} - 4.$$

Final answer

$$a = 4 + 4\sqrt{5}, \quad b = 4\sqrt{5} - 4.$$

Question 5

Trigonometric Equations

5. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Solve, for $-180^\circ < \theta \leq 180^\circ$, the equation

$$3 \tan(\theta + 43^\circ) = 2 \cos(\theta + 43^\circ)$$

(6)

Worked Solution - Question 5

1. Substitute u

Let $u = \theta + 43^\circ$. The equation becomes $3 \tan u = 2 \cos u$.

2. Clear the cosine denominator

$$3 \frac{\sin u}{\cos u} = 2 \cos u, \text{ so } 3 \sin u = 2 \cos^2 u.$$

3. Use cos squared

$$3 \sin u = 2(1 - \sin^2 u), \text{ so } 2 \sin^2 u + 3 \sin u - 2 = 0.$$

4. Factor the quadratic

$$(2 \sin u - 1)(\sin u + 2) = 0.$$

5. Use the valid sine value

$$\sin u = -2 \text{ is impossible, so } \sin u = \frac{1}{2}.$$

6. Find u in the interval

Since $-180^\circ < \theta \leq 180^\circ$, $-137^\circ < u \leq 223^\circ$. Thus $u = 30^\circ$ or 150° .

7. Return to theta

$$\theta = u - 43^\circ, \text{ so } \theta = -13^\circ \text{ or } 107^\circ.$$

Final answer

$$\theta = -13^\circ, 107^\circ.$$

Question 6

Geometric Sequences

6. In a geometric sequence $u_1, u_2, u_3, \dots$

- the common ratio is r
- $u_2 + u_3 = 6$
- $u_4 = 8$

(a) Show that r satisfies

$$3r^2 - 4r - 4 = 0 \quad (3)$$

Given that the geometric sequence has a sum to infinity,

(b) find u_1 (3)

(c) find S_∞ (2)

Worked Solution - Question 6

1. Write the terms

$$u_2 = u_1 r, u_3 = u_1 r^2, \text{ and } u_4 = u_1 r^3.$$

2. Use u four

$$\text{Since } u_4 = 8, u_1 r^3 = 8, \text{ so } u_1 r = \frac{8}{r^2}.$$

3. Use u two plus u three

$$u_2 + u_3 = u_1 r(1 + r) = 6.$$

4. Show the quadratic

$$\text{Substitute } u_1 r = \frac{8}{r^2} \text{ to get } \frac{8}{r^2}(1 + r) = 6. \text{ Hence } 8 + 8r = 6r^2, \text{ so } 3r^2 - 4r - 4 = 0.$$

5. Solve for r

$$3r^2 - 4r - 4 = 0 \text{ gives } r = 2 \text{ or } r = -\frac{2}{3}.$$

6. Choose the sum-to-infinity ratio

$$\text{A sum to infinity requires } |r| < 1, \text{ so } r = -\frac{2}{3}.$$

7. Find u one

$$u_1 r^3 = 8, \text{ so } u_1 \left(-\frac{2}{3}\right)^3 = 8 \text{ and } u_1 = -27.$$

8. Find the infinite sum

$$S_\infty = \frac{u_1}{1 - r} = \frac{-27}{1 + 2/3} = -\frac{81}{5}.$$

Final answer

(a) Proven: $3r^2 - 4r - 4 = 0$.

(b) $u_1 = -27$. (c) $S_\infty = -\frac{81}{5}$.

Question 7

Integration

7. $f(x) = Ax^3 + 6x^2 - 4x + B$

where A and B are constants.

Given that

- $(x + 2)$ is a factor of $f(x)$
- $\int_3^5 f(x) dx = 176$

find the value of A and the value of B .

(7)

Worked Solution - Question 7

1. Use the factor condition

Since $(x + 2)$ is a factor, $f(-2) = 0$.

2. Form the first equation

$A(-2)^3 + 6(-2)^2 - 4(-2) + B = 0$, so $-8A + 32 + B = 0$ and $B = 8A - 32$.

3. Set up the integral

$$\int_3^5 (Ax^3 + 6x^2 - 4x + B) dx = 176.$$

4. Integrate

$$\left[\frac{A}{4}x^4 + 2x^3 - 2x^2 + Bx \right]_3^5 = 176.$$

5. Simplify the second equation

This gives $136A + 164 + 2B = 176$, so $68A + B = 6$.

6. Solve simultaneously

Using $B = 8A - 32$ in $68A + B = 6$ gives $76A = 38$, so $A = \frac{1}{2}$.

7. Find B

$$B = 8\left(\frac{1}{2}\right) - 32 = -28.$$

Final answer

$$A = \frac{1}{2}, \quad B = -28.$$

Question 8**Applications of Differentiation**

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A curve has equation

$$y = 256x^4 - 304x - 35 + \frac{27}{x^2} \quad x \neq 0$$

(a) Find $\frac{dy}{dx}$

(3)

(b) Hence find the coordinates of the stationary points of the curve.

(5)

Worked Solution - Question 8

Topic group

1. Rewrite with powers

$$y = 256x^4 - 304x - 35 + 27x^{-2}.$$

2. Differentiate

$$\frac{dy}{dx} = 1024x^3 - 304 - 54x^{-3} = 1024x^3 - 304 - \frac{54}{x^3}.$$

3. Set the derivative to zero

$$1024x^3 - 304 - \frac{54}{x^3} = 0.$$

4. Use a substitution

Multiplying by x^3 gives $1024x^6 - 304x^3 - 54 = 0$. Let $u = x^3$.

5. Solve for u

$$1024u^2 - 304u - 54 = 0, \text{ or } 512u^2 - 152u - 27 = 0 = (8u + 1)(64u - 27).$$

6. Find x

$$\text{Thus } x^3 = -\frac{1}{8} \text{ or } x^3 = \frac{27}{64}, \text{ giving } x = -\frac{1}{2} \text{ or } x = \frac{3}{4}.$$

7. Find the y values

$$\text{Substitution into } y = 256x^4 - 304x - 35 + \frac{27}{x^2} \text{ gives } y = 241 \text{ at } x = -\frac{1}{2} \text{ and } y = -134 \text{ at } x = \frac{3}{4}.$$

8. State the stationary points

$$\text{The stationary points are } \left(-\frac{1}{2}, 241\right) \text{ and } \left(\frac{3}{4}, -134\right).$$

Final answer

$$(a) \frac{dy}{dx} = 1024x^3 - 304 - \frac{54}{x^3}. \quad (b) \left(-\frac{1}{2}, 241\right), \left(\frac{3}{4}, -134\right).$$

Question 9

Modelling with Sequences & Series

9. A scientist is using carbon-14 dating to determine the age of some wooden items.

The equation for carbon-14 dating an item is given by

$$N = k\lambda^t$$

where

- N grams is the amount of carbon-14 **currently** present in the item
- k grams was the **initial** amount of carbon-14 present in the item
- t is the number of years since the item was made
- λ is a constant, with $0 < \lambda < 1$

- (a) Sketch the graph of N against t for $k = 1$

(2)

Given that it takes 5700 years for the amount of carbon-14 to reduce to half its initial value,

- (b) show that the value of the constant λ is 0.999878 to 6 decimal places.

(2)

Given that Item A

- is known to have had 15 grams of carbon-14 present initially
- is thought to be 3250 years old

- (c) calculate, to 3 significant figures, how much carbon-14 the equation predicts is currently in Item A .

(2)

Item B is known to have initially had 25 grams of carbon-14 present, but only 18 grams now remain.

- (d) Use algebra to calculate the age of Item B to the nearest 100 years.

(3)

Worked Solution - Question 9

Topic group

1. Sketch the model for k equals one

For $k = 1$, $N = \lambda^t$ with $0 < \lambda < 1$. The graph starts at $(0, 1)$, decreases as t increases, and tends towards 0.

2. Use the half-life information

After 5700 years, $N = \frac{k}{2}$, so $k\lambda^{5700} = \frac{k}{2}$.

3. Find lambda

$\lambda^{5700} = \frac{1}{2}$, hence $\lambda = \left(\frac{1}{2}\right)^{1/5700} = 0.999878\dots$, so $\lambda = 0.999878$ to 6 decimal places.

4. Model item A

For item A, $k = 15$ and $t = 3250$, so $N = 15\lambda^{3250}$.

5. Calculate item A

$N = 15(0.9998784026\dots)^{3250} = 10.103\dots$, so the current amount is 10.1 g to 3 significant figures.

6. Set up item B

For item B, $18 = 25\lambda^t$, so $\lambda^t = \frac{18}{25}$.

7. Take logarithms

$t \log \lambda = \log \left(\frac{18}{25} \right)$, so $t = \frac{\log(18/25)}{\log \lambda}$.

8. Round the age

Using $\lambda = (1/2)^{1/5700}$ gives $t = 2701 \dots$, so the age is **2700** years to the nearest 100 years.

Final answer

(a) Decreasing exponential through $(0, 1)$.

(b) $\lambda = 0.999878$.

(c) 10.1 g. (d) 2700 years.

Question 10

10. The circle C has centre $X(3, 5)$ and radius r

The line l has equation $y = 2x + k$, where k is a constant.

(a) Show that l and C intersect when

$$5x^2 + (4k - 26)x + k^2 - 10k + 34 - r^2 = 0 \quad (3)$$

Given that l is a tangent to C ,

(b) show that $5r^2 = (k + p)^2$, where p is a constant to be found. (3)

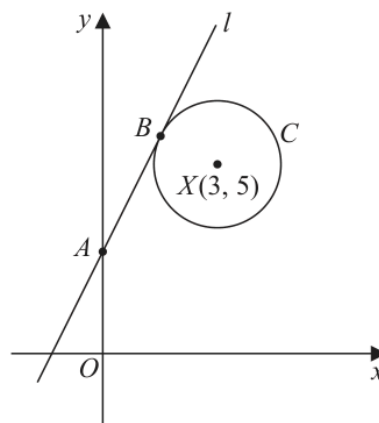


Figure 2

The line l

- cuts the y -axis at the point A
- touches the circle C at the point B

as shown in Figure 2.

Given that $AB = 2r$

(c) find the value of k (6)

(Total 12 marks)

Worked Solution - Question 10

1. Write the circle equation

The circle has centre $(3, 5)$ and radius r , so $(x - 3)^2 + (y - 5)^2 = r^2$.

2. Substitute the line

On the line, $y = 2x + k$. Substitution gives $(x - 3)^2 + (2x + k - 5)^2 = r^2$.

3. Expand

$$x^2 - 6x + 9 + 4x^2 + 4kx - 20x + k^2 - 10k + 25 - r^2 = 0.$$

4. Show the intersection equation

Collecting terms gives $5x^2 + (4k - 26)x + k^2 - 10k + 34 - r^2 = 0$.

5. Use tangency

For a tangent, the quadratic has one repeated root, so its discriminant is zero.

6. Set the discriminant to zero

$$(4k - 26)^2 - 20(k^2 - 10k + 34 - r^2) = 0.$$

7. Simplify

This reduces to $20r^2 - 4k^2 - 8k - 4 = 0$, hence $5r^2 = (k + 1)^2$. Therefore $p = 1$.

8. Find the tangent point x coordinate

For the repeated root, $x_B = -\frac{4k - 26}{2 \cdot 5} = \frac{13 - 2k}{5}$.

9. Use distance AB

The point A is $(0, k)$ and B lies on a line of gradient 2, so

$$AB = \sqrt{x_B^2 + (2x_B)^2} = \sqrt{5} x_B.$$

10. Use the radius relation

From $5r^2 = (k + 1)^2$, $r = \frac{k + 1}{\sqrt{5}}$ in the shown configuration.

11. Apply AB equals two r

$$\sqrt{5} \left(\frac{13 - 2k}{5} \right) = 2 \left(\frac{k + 1}{\sqrt{5}} \right).$$

12. Solve for k

Multiplying by $\sqrt{5}$ gives $13 - 2k = 2k + 2$, so $k = \frac{11}{4}$.

Final answer

(a) Proven.

(b) $5r^2 = (k + 1)^2$, so $p = 1$.

(c) $k = \frac{11}{4}$.